HP / ENGINEERS INDIA LIMITED (A Govt. of India Undertaking)	**Residue Upgradation Facility (RUF)** **EPCC-3 Package for** **Visakh Refinery Modernization Project (VRMP)**	L&T Hydrocarbon Engineering **L&T- CHIYODA LIMITED**
Title: ~~Data Sheet With~~ GAD for Dock Leveller (504-LZ-930 A/B) **Doc. No.: B016-RUF-LT-504-MP-GN-NTPL(1)-13002**		**Rev. No.: 00** **Page No: 1 of 4**

OWNER	:	**HINDUSTAN PETROLEUM CORPORATION LIMITED (HPCL)**
PMC	:	**ENGINEERS INDIA LIMITED, NEW DELHI (EIL)**
UNIT	:	**504**
PMC JOB NO.	:	**B016**
EQUIPMENT DESCRIPTION	:	**DOCK LEVELLER**
EQUIPMENT TAG	:	504-LZ-930 A/B
P. R. NO.	:	B016-RUF-LT-504-MP-PR-0014
VENDOR NAME	:	**NIHVA TECHNOLOGIES PVT LTD**
VENDOR DOCUMENT NO.	:	**NIHVA_20-21_0157**

DOCUMENT SUBMISSION STATUS – LTHE/VENDOR	**REVIEW STATUS**	
	LTHE	**EIL**
DOC. CATEGORY: (USE TICK MARK) ☑ R ☐ I	☐ Code 1 – No Comments. Proceed With Manufacture / Fabrication / Construction As Per The Document.	☐ Code 1 – No Comments. Proceed With Manufacture / Fabrication / Construction As Per The Document.
ISSUED FOR ☐ INFORMATION ☑ REVIEW ☐ CONSTRUCTION ☐ AS BUILT	☑ Code 2 – Proceed With Manufacture/Fabrication/Construction As Per Commented Document. Revised Document Required. ☐ Review Code 3 – Document Does Not Conform To Basic Requirements As Marked. Resubmit For Review ☐ Code R– Document Is Retained For Records. Proceed With Manufacture / Fabrication ☐ Code V- Void (Submission Required / Not Required)	☐ Code 2 – Proceed With Manufacture/Fabrication/Construction As Per Commented Document. Revised Document Required. ☐ Review Code 3 – Document Does Not Conform To Basic Requirements As Marked. Resubmit For Review ☐ Code R– Document Is Retained For Records. Proceed With Manufacture / Fabrication ☐ Code V- Void (Submission Required / Not Required)
	ME-P / DISCIPLINE | MJG/ PPP/ MRS / SIGN | 04-Jun-21 / DATE	DISCIPLINE | SIGN | DATE

Revision No.	Description	Prepared by	Reviewed by	Approved by	Reviewed by	Approved by	Approved Date
		Vikrant J	Amit K	Amit K			31-05-2021

DOCUMENT REVIEW DETAILS

1 No Comments.
2 Proceed with manufacture / fabrication as per the commented document. Revised document required.
3 Document does not conform to basic requirements as marked. Resubmit for review.
R Retained for Records. V Void.

Signature with Date: 03-SEP-2021
Name of Reviewer: AJEET KUMAR (B565)
Page No. 1 of 19

Doc. B016-RUF-LT-504-MP-GN-NTPL(1)-13002 Rev: 00

ALL RIGHTS RESERVED
THIS DOCUMENT AND ANY DATA AND INFORMATION CONTAINED THEREIN ARE CONFIDENTIAL AND THE PROPERTY OF L&T HYDROCARBON ENGINEERING (LTHE) AND THE COPYRIGHT THEREIN IS VESTED IN LTHE NO PART OF THIS DOCUMENT, DATA, OR INFORMATION SHALL BE DISCLOSED TO OTHERS OR REPRODUCED, STORED IN A RETRIEVAL SYSTEM OR USED FOR ANY PURPOSE WHATSOEVER, EXCEPT WITH THE PRIOR WRITTEN PERMISSION OF LTHE.

REVISION HISTORY SHEET

REV. NO.	REV. DATE	REVISED PAGE	REVISION DESCRIPTION
00	31-05	NA	ISSUED FOR APPROVAL

DOCUMENT REVIEW DETAILS

1 No Comments. **Doc. B016-RUF-LT-504-MP-GN-NTPL(1)-13002 Rev. 00**

2 ✓ Proceed with manufacture / fabrication as per the commented document. Revised document required.

3 Document does not conform to basic requirements as marked. Resubmit for review.

R Retained for Records. V Void.

Signature with Date **03-SEP-2021**

Name of Reviewer **AJEET KUMAR (B565)**

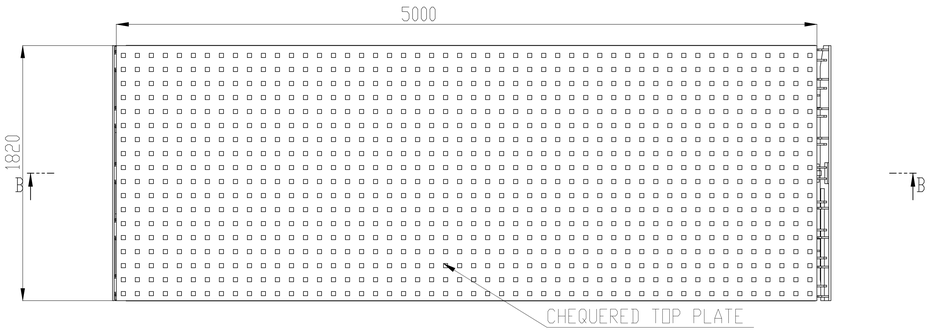
Page No. **2 of 19**

ALL RIGHTS RESERVED

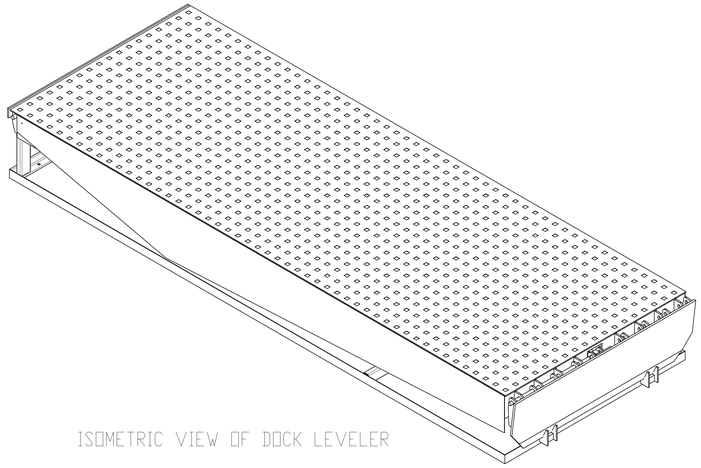
THIS DOCUMENT AND ANY DATA AND INFORMATION CONTAINED THEREIN ARE CONFIDENTIAL AND THE PROPERTY OF L&T HYDROCARBON ENGINEERING (LTHE) AND THE COPYRIGHT THEREIN IS VESTED IN LTHE NO PART OF THIS DOCUMENT, DATA, OR INFORMATION SHALL BE DISCLOSED TO OTHERS OR REPRODUCED, STORED OR USED FOR ANY PURPOSE WHATSOEVER, EXCEPT WITH THE PRIOR WRITTEN PERMISSION OF LTHE.

ALL RIGHTS RESERVED
THIS DOCUMENT AND ANY DATA AND INFORMATION CONTAINED THEREIN ARE CONFIDENTIAL AND THE PROPERTY OF L&T-CHIYODA LIMITED (L&T-CHIYODA), AND SHALL NOT BE DISCLOSED TO OTHERS OR REPRODUCED IN ANY MANNER WITHOUT THE EXPRESS WRITTEN PERMISSION OF L&T-CHIYODA. IF THIS DOCUMENT IS LOANED OR DISCLOSED TO OTHERS OR REPRODUCED IN ANY MANNER OR USED FOR ANY PURPOSE WHATSOEVER, L&T-CHIYODA SHALL BE ENTITLED TO TAKE ALL STEPS AS IT DEEMS FIT AND PROPER.

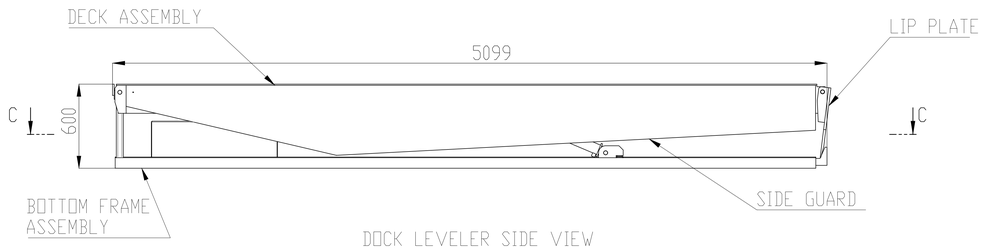
DRG. NO.



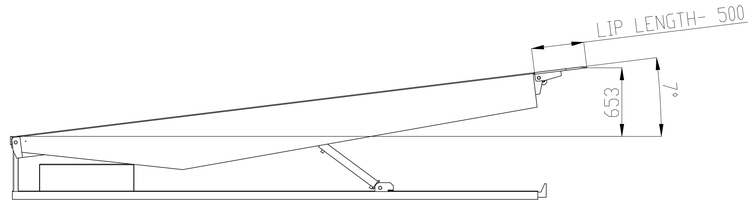
DOCK LEVELER TOP VIEW



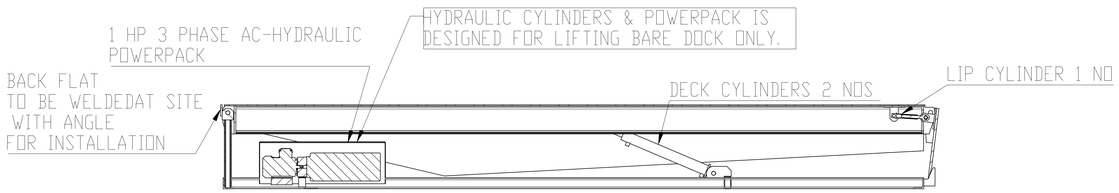
ISOMETRIC VIEW OF DOCK LEVELER



DOCK LEVELER SIDE VIEW



DECK UP HEIGHT - 600 MM



SECTION B-B
SCALE 1 : 15

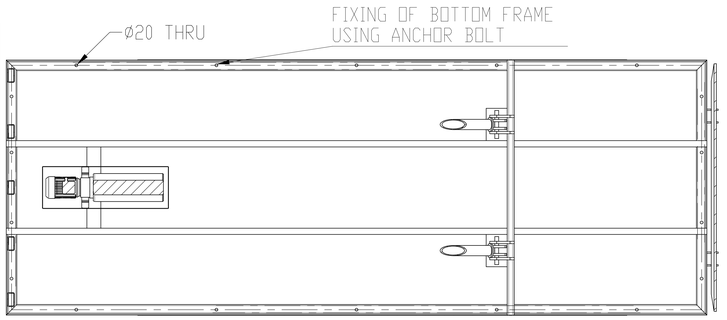


DECK DOWN HEIGHT - 300 MM

TECHNICAL SPECIFICATIONS OF PIT MOUNTED ELECRO-HYDRAULIC DOCK LEVELER -

1. SIZE OF DOCK LEVELER - 1820mm(W) X 5000mm(W)
2. OVERALL SIZE OF PRODUCT - 1820mm(W)X5100mm(L)X600mm(H)
3. CAPACITY - 6 TON POINT LOAD
4. LIP PLATE SIZE - 500mm(L)X1820mm(W)
5. DECK PLATE - CHEQUERED SHEET 1820mm(W)x5000mm(L)
6. DECK & LIP UP-DOWN OPERATION BY HYDRAULIC CYLINDER
7. DECK UP HEIGHT-(+) 600 MM
8. DECK DOWN HEIGHT-(-) 300MM
9. HEIGHT OF DOCK LEVELER FROM ROAD LEVEL- 900 mm ,
10. OPERATION CONTROL BY PUSH BUTTONS ON CONTROL PANEL MOUNTED ON STAND NEAR DOCK LEVELER
11. ELECTRICAL SUPPLY OF EQUIPMENT - 415 VOLT 3 PHASE AC , 50HZ
12. PAINT - FIROZI BLUE
13. MATERIAL OF CONSTRUCTION - MILD STEEL AS PER IS STANDARD
14. MANUFACTURING STANDARD OF DOCK LEVELER -EN1398
15. ELECRICAL MOTOR COMPLIANCE ZONE 1-IIC-T3 HAZARDOUS AREA . FOR OTHER EQUIPMENT-PESO CERTIFICATION
16. PIT SIZE - 1870 mm(W)X5100mm(L)x600 mm(D)
17. DOCK LEVELER FIXING IN PIT BY ANCHOR BOLT .
18. REQUIRED PIT & ALL CIVIL WORK FOR MOUNTING OD DOCK LEVELER IS IN CLIENT SCOPE .

REFER PIT DRAWING FOR CONSTRUCTION OF PIT .



SECTION C-C
SCALE 1 : 15

DOCUMENT REVIEW DETAILS

1 No Comments.
2 Proceed with manufacture / fabrication as per the commented document. Revised document required.
3 Document does not conform to basic requirements as marked. Resubmit for review.
R Retained for Records. V Void.

Signature with Date: 03-SEP-2021
Name of Reviewer: AJEET KUMAR (8580)
Page No.: 3 of 19

Doc. BGS-RUF-LT-504-MP-GN-NTFL(1)-13002 Rev. 00

Provide reference document number

L&T - CHIYODA LIMITED

- ☐ REVIEW CODE 1 – NO COMMENTS PROCEED WITH MANUFACTURE / FABRICATION/ CONSTRUCTION AS PER THE DOCUMENT.
- ☐ REVIEW CODE 2 – PROCEED WITH MANUFACTURE /FABRICATION /CONSTRUCTION AS PER THE COMMENTED DOCUMENT. REVISED DOCUMENT REQUIRED.
- ☐ REVIEW CODE 3 – DOCUMENT DOES NOT CONFORM TO BASIC REQUIREMENTS AS MARKED. RESUBMIT FOR REVIEW.
- ☐ R – DOCUMENT IS RETAINED FOR RECORDS. PROCEED WITH MANUFACTURE / FABRICATION
- ☐ V–VOID

COMMENTED BY	REVIEWED BY	APPROVED BY	DATE

THIS REVIEW / APPROVAL DOES NOT RELIEVE THE VENDOR / MANUFACTURER OF THE FULL RESPONSIBILITY AS TO DESIGN CORRECTNESS OF DIMENSIONS & COMPLIANCE WITHALL APPLICABLE REGULATIONS AND WITH THE REQUIREMENTS MADE IN PURCHASER SPECIFICATIONS.

DRAWING NO.	DRAWING TITLE	REV
	REFERENCE DOCUMENTS / DRAWINGS	

DRG. CATEGORY: (USE TICK MARK) R I
EX. REVIEW/APPROVAL CODE:
VPTL NO.:
SCH. DATE OF SUBMISSION:
DOCUMENT STATUS
STATEMENT OF SUB: 1st 2nd 3rd
☐ FINAL FOR CODE-1 ☑ AS BUILT
ISSUED FOR
☐ AS BUILT ☐ INFORMATION
☐ CONSTRUCTION ☐ REVIEW
SIGN: DATE:

- ☐ REVIEW CODE 1 – NO COMMENTS PROCEED WITH MANUFACTURE / FABRICATION/ CONSTRUCTION AS PER THE DOCUMENT.
- ☐ REVIEW CODE 2 – PROCEED WITH MANUFACTURE / FABRICATION / CONSTRUCTION AS PER THE COMMENTED DOCUMENT. REVISED DOCUMENT REQUIRED
- ☐ REVIEW CODE 3 – DOCUMENT DOES NOT CONFORM TO BASIC REQUIREMENTS AS MARKED. RESUBMIT FOR REVIEW.
- ☐ R – DOCUMENT IS RETAINED FOR RECORDS. PROCEED WITH MANUFACTURE / FABRICATION
- ☐ V–VOID

NAME	DISCIPLINE	SIGN.	DATE

REV.	DESCRIPTION	DRN	CHECKED	APPROVED	REVIEWED	APPROVED	DATE
00	ISSUED FOR APPROVAL	SMS	VJ	AK			

OWNER: HINDUSTAN PETROLEUM CORPORATION LIMITED
VISAKH REFINERY MODERNIZATION PROJECT
EPCC-03 PACKAGE FOR RESIDUE UP-GRADATION FACILITY (RUF)

PMC : ENGINEERS INDIA LIMITED (A Govt. of India Undertaking)

LSTK CONTRACTOR: L&T Hydrocarbon Engineering

DETAILED ENGINEERING CENTER: L&T - CHIYODA LIMITED

VENDOR :	VENDOR PO NO:
NIHVA TECHNOLOGIES PVT LTD	REFINARY/75000-82256/VMM
	VENDOR DWG. NO: NIHVA_20-21_0157

DATE	SCALE	TITLE:
31.05.21	1:1	DATASHEET WITH GAD FOR DOCK LEVELER

DEPT:	PROJECT:	DRG. NO.:	SH.1 OF 1
DESIGN	L&T-RUF	BGS-RUF-LT-504-MP-GN-NTFL(1)-13002	
DRN BY: SMS	PWO NO.:		REV. 00
CHK BY: VJ			
APP BY: AK	CAD FILE: BGS RG DRL GSNC GD SYSTEM DWT		

DATA SHEET FOR DOCK LEVELLER –

Sr No	Parameter	Remark
1	Required Capacity	06 Ton
2	Leveller operation mode	AC power pack hydraulically operated
3	Lip operation mode	hydraulic
4	Operation control	by push buttons and control panel
5	Loading platform height from road Level	900 mm
6	Travel above & below dock level	600 mm & 300 mm
7	Platform Size (W x L)	1820 mm x 5000 mm
8	Lip size (W x L)	1820 mm x 500 mm
9	Maximum slope can be achieved	7 Degree

* Please refer Attachment-1 for standard catalog of Dock Leveller.

DOCUMENT REVIEW DETAILS

1 No Comments. **Doc. B016-RUF-LT-504-MP-GN-NTPL(1)-13002 Rev. 00**

2 ✓ Proceed with manufacture / fabrication as per the commented document. Revised document required.

3 Document does not conform to basic requirements as marked. Resubmit for review.

R Retained for Records. V Void.

Signature with Date **03-SEP-2021**

Name of Reviewer **AJEET KUMAR (B565)**

Page No. **4 of 19**

ALL RIGHTS RESERVED
THIS DOCUMENT AND ANY DATA AND INFORMATION CONTAINED THEREIN ARE CONFIDENTIAL AND THE PROPERTY OF L&T HYDROCARBON ENGINEERING (LTHE) AND THE COPYRIGHT THEREIN IS VESTED IN LTHE NO PART OF THIS DOCUMENT, DATA, OR INFORMATION SHALL BE DISCLOSED TO OTHERS OR REPRODUCED, STORED OR USED FOR ANY PURPOSE WHATSOEVER, EXCEPT WITH THE PRIOR WRITTEN PERMISSION OF LTHE.

ATTACHMENT-1

NIHVA

NIHVA TECHNOLOGIES PVT. LTD.

Dock Leveller



- SOLUTIONS
- SERVICES
- SATISFACTION

DOCUMENT REVIEW DETAILS

1 No Comments. **Doc. B016-RUF-LT-504-MP-GN-NTPL(1)-13002 Rev. 00**

2 ✓ Proceed with manufacture / fabrication as per the commented document. Revised document required.

3 Document does not conform to basic requirements as marked. Resubmit for review.

R Retained for Records. V Void.

Signature with Date **03-SEP-2021**

Name of Reviewer **AJEET KUMAR (B565)**

Page No. **5 of 19**

www.nihva.com

LOADING EQUIPMENTS

NIHVA Technologies Pvt.Ltd is the premier solutions provider of Safe, Effective, and technologically innovative material handling equipment's.

We offer verity of Loading Equipments including Dock levellers, Dock Shelters, Dock Seal, Scissor Lifts, Good Lifts along with other material handling products. We offer turnkey solutions for the most demanding freight-handling needs. Dock Levellers are one of the simplest equipment, designed according to the Indian conditions. Continues operations & ruggedness for not only helping the loading & unloading the vehicles easily but also reducing the dispatch time of the vehicles significantly.

Our gamut of hydraulic dock levellers design & fabricated extensive used for heavy loading and unloading.





FEATURES

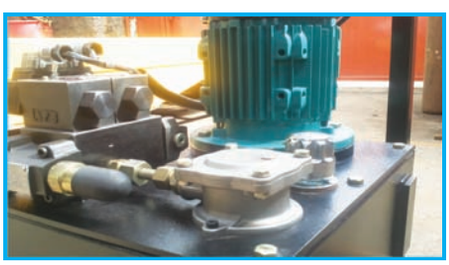
- Robust design, safe operations and long term reliability.
- Total control on basic hydraulic like pressure and flow.
- Velocity safety stops.
- Simple electric panel with push button.
- Priority to safe working operations.
- Long excellent stability.
- Consistency in working performance.

DOCUMENT REVIEW DETAILS
1 No Comments.
2 Proceed with manufacture / fabrication as per the commented document. Revised document required.
3 Document does not conform to design requirements as marked. Resubmit for review.
R Retained for Records.
Signature with Date 03-SEP-2021
Name of Reviewer AJEET KUMAR (B565)
Page No. 6 of 19
Doc. B016-RUF-LT-504-MP-GN-NTPL(1)-13002 Rev. 00



Why NIHVA dock leveller?

- Heavy duty structure to take the dynamic load.



- Stronger Base Frame with self-standing capability.
- Stronger Lip & rear Hinges.



- 16 mm thick lip plate with step at tip for adopting smaller vehicles.
- Rear Strong hinge support for self standing capacity.
- Collapsible Telescopic Side Guard for safety protection.



- Single Approachable steady rest for maintenance.
- Stand alone power pack and control panel for ease of operation and maintenance.

DOCUMENT REVIEW DETAILS

1 No Comments. **Doc. B016-RUF-LT-504-MP-GN-NTPL(1)-13002 Rev. 00**

2 ✓ Proceed with manufacture / fabrication as per the commented document. Revised document required.

3 Document returned with comments, requirements as marked. Resubmit for review.

R Retained for records. V Void.

Signature with Date **03-SEP-2021**

Name of Reviewer **AJEET KUMAR (B565)**

Page No. **7 of 19**

SERVICE BACK UP / MAINTENANCE / AMC



NIHVA SERVICE SOLUTIONS

Preventing a problem is always better than fixing one. That simple philosophy is behind all our Service Contracts as they are designed to reduce operational risks and help provide you with a trouble-free materials handling operation. This allows you to concentrate on your core business rather than worrying about unexpected breakdowns, scheduling maintenance, ordering parts or managing service engineers.

MULTIPLE OF SERVICES & AMC SOLUTIONS

Simply choose the Service Contract you require and let us do the rest:

- **SAFETY INSPECTION**

  Designed to keep your Equipment's operating safely and provide protection for your workforce. This contract ensures that your Equipment's always comply with the safety standards set by national laws and regulating bodies.

- **PREVENTIVE MAINTENANCE**

  This contract is an effective way to prevent unplanned downtime and costly repairs. We take responsibility for maintaining your Equipment according to a pre-determined plan as well as ensuring that they meet all safety requirements.

- **FULL SERVICE**

  This contract is designed to take care of all your equipment's maintenance and repair requirements in order to provide you with the highest levels of operational performance and uptime.

  We offer the services in order to customize your contract as per your exact needs: 24/7 service, On-site NIHVA service technician, entire NIHVA maintenance team along with combined support package.

OUR ACTIVITIES

- Door Automation
- Automation Solutions
- Material Handling Equipments

DOCUMENT REVIEW DETAILS
1 No Comments.
2 Proceed with manufacture / fabrication as per the commented document. Revised document required.
3 Document does not conform to the requirements as marked. Resubmit for review.
R Retained for Records. V Void.
Signature with Date 03-SEP-2021
Name of Reviewer AJEET KUMAR (B565)
Page No. 8 of 19
Doc. B016-RUF-LT-504-MP-GN-NTPL(1)-13002 Rev. 00

OPTIONAL ACCESSORIES

Rubber Bumper



Flashing Light



Traffic Light



Dock Lights



Vehicle Wheel Restraint



Vehicle Wheel Alignment guides



NIHVA TECHNOLOGIES PVT. LTD.

Regd. Office:
8/1 Coromant Hsg. Soc. Sec. No. 27,
Nigdi, Pradhikaran, Pune - 411044.

Tel. : + 91 20 6630 4033
Mob. : + 91 855 109 1333
Email: sales@nihva.com
Web. : www.nihva.com

NIHVA

DOCUMENT REVIEW DETAILS

1 No Comments.
2 Proceed with manufacture/fabrication as per the commented document. Revised document required.
3 Document does not conform to basic requirements as marked. Resubmit for review.
R Retained for Records.
V Void.

Signature with Date 08-SEP-2021
Name of Reviewer AJEET KUMAR (B565)
Page No. 9 of 19

Doc. B016-RUF-LT-504-MP-GN-NTPL(1)-13002 Rev. 00

Refer attached commented drawing.

DOCUMENT REVIEW DETAILS

1 No Comments. **Doc. B016-RUF-LT-504-MP-GN-NTPL(1)-13002 Rev. 00**

2 ✓ Proceed with manufacture / fabrication as per the commented document. Revised document required.

3 Document does not conform to basic requirements as marked. Resubmit for review.

R Retained for Records. V Void.

Signature with Date **03-SEP-2021**

Name of Reviewer **AJEET KUMAR (B565)**

Page No. **10 of 19**

OWNER	:	HINDUSTAN PETROLEUM CORPORATION LIMITED (HPCL)
PMC	:	ENGINEERS INDIA LIMITED, NEW DELHI (EIL)
UNIT	:	504
PMC JOB NO.	:	B016
EQUIPMENT DESCRIPTION	:	DOCK LEVELLER
EQUIPMENT TAG	:	504-LZ-930 A/B
P. R. NO.	:	B016-RUF-LT-504-MP-PR-0014
VENDOR NAME	:	NIHVA TECHNOLOGIES PVT LTD
VENDOR DOCUMENT NO.	:	NIHVA_20-21_0157

DOCUMENT SUBMISSION STATUS – LTHE/VENDOR	REVIEW STATUS – LTHE	REVIEW STATUS – EIL
DOC. CATEGORY: (USE TICK MARK) ☑ R ☐ I	☐ Code 1 – No Comments. Proceed With Manufacture / Fabrication / Construction As Per The Document.	☐ Code 1 – No Comments. Proceed With Manufacture / Fabrication / Construction As Per The Document.
ISSUED FOR ☐ INFORMATION ☑ REVIEW ☐ CONSTRUCTION ☐ AS BUILT	☑ Code 2 – Proceed With Manufacture/Fabrication/Construction As Per Commented Document. Revised Document Required. ☐ Review Code 3 – Document Does Not Conform To Basic Requirements As Marked. Resubmit For Review ☐ Code R– Document Is Retained For Records. Proceed With Manufacture / Fabrication ☐ Code V- Void (Submission Required / Not Required)	☐ Code 2 – Proceed With Manufacture/Fabrication/Construction As Per Commented Document. Revised Document Required. ☐ Review Code 3 – Document Does Not Conform To Basic Requirements As Marked. Resubmit For Review ☐ Code R– Document Is Retained For Records. Proceed With Manufacture / Fabrication ☐ Code V- Void (Submission Required / Not Required)
	ME-P / DISCIPLINE | MJG/ PPP/ MRS / SIGN | 04-Jun-21 / DATE	DISCIPLINE | SIGN | DATE

Revision No.	Description	Prepared by	Reviewed by	Approved by	Reviewed by	Approved by	Approved Date
	DOCUMENT REVIEW DETAILS	Vikrant J	Amit K	Amit K			31-05-2021

1 No Comments.
2 Proceed with manufacture / fabrication as per the commented document. Revised document required.
3 Document does not conform to basic requirements as marked. Resubmit for review.
R Retained for Records. V Void.

Signature with Date: 03-SEP-2021
Name of Reviewer: AJEET KUMAR (B565)
Page No. 11 of 19

Doc. B016-RUF-LT-504-MP-GN-NTPL(1)-13002 Rev: 00

ALL RIGHTS RESERVED
THIS DOCUMENT AND ANY DATA AND INFORMATION CONTAINED THEREIN ARE CONFIDENTIAL AND THE PROPERTY OF L&T HYDROCARBON ENGINEERING (LTHE) AND THE COPYRIGHT THEREIN IS VESTED IN LTHE NO PART OF THIS DOCUMENT, DATA, OR INFORMATION SHALL BE DISCLOSED TO OTHERS OR REPRODUCED, STORED IN A RETRIEVAL SYSTEM OR USED FOR ANY PURPOSE WHATSOEVER, EXCEPT WITH THE PRIOR WRITTEN PERMISSION OF LTHE.

REVISION HISTORY SHEET

REV. NO.	REV. DATE	REVISED PAGE	REVISION DESCRIPTION
00	31-05	NA	ISSUED FOR APPROVAL

DOCUMENT REVIEW DETAILS

1 No Comments. **Doc. B016-RUF-LT-504-MP-GN-NTPL(1)-13002 Rev. 00**

2 ✓ Proceed with manufacture / fabrication as per the commented document. Revised document required.

3 Document does not conform to basic requirements as marked. Resubmit for review.

R Retained for Records. V Void.

Signature with Date **03-SEP-2021**

Name of Reviewer **AJEET KUMAR (B565)**

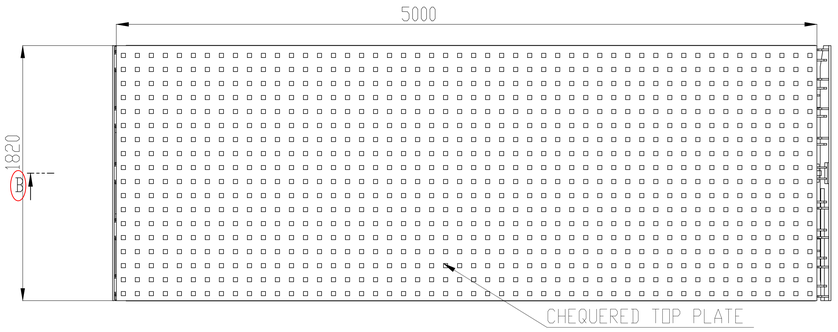
Page No. **12 of 19**

ALL RIGHTS RESERVED

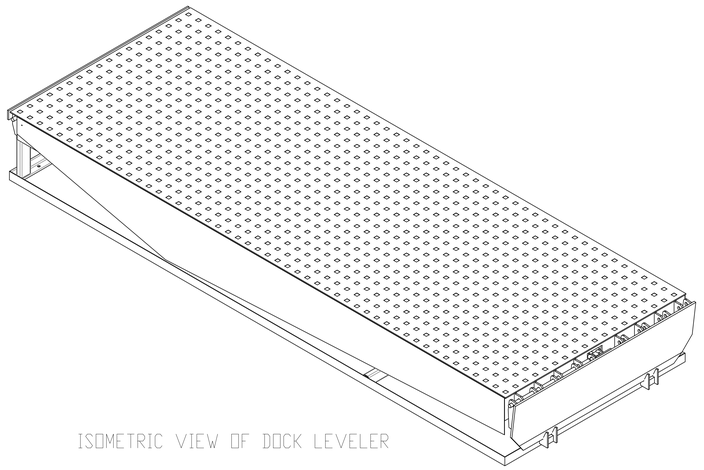
THIS DOCUMENT AND ANY DATA AND INFORMATION CONTAINED THEREIN ARE CONFIDENTIAL AND THE PROPERTY OF L&T HYDROCARBON ENGINEERING (LTHE) AND THE COPYRIGHT THEREIN IS VESTED IN LTHE NO PART OF THIS DOCUMENT, DATA, OR INFORMATION SHALL BE DISCLOSED TO OTHERS OR REPRODUCED, STORED OR USED FOR ANY PURPOSE WHATSOEVER, EXCEPT WITH THE PRIOR WRITTEN PERMISSION OF LTHE.

ALL RIGHTS RESERVED
THIS DOCUMENT AND ANY DATA AND INFORMATION CONTAINED THEREIN ARE CONFIDENTIAL AND THE PROPERTY OF L&T-CHIYODA LIMITED (LTCH), AND ARE SUBMITTED ON THE CONDITION THAT THEY WILL NOT BE REPRODUCED, USED, DISCLOSED TO OTHERS OR DISPOSED OF IN ANY MANNER WITHOUT THE PRIOR WRITTEN CONSENT OF LTCH. BY ACCEPTING THIS DOCUMENT, THE RECIPIENT AGREES THAT THE INFORMATION SHALL BE DISCLOSED TO OTHERS OR REPRODUCED IN ANY MANNER WHATSOEVER WITHOUT THE PRIOR WRITTEN PERMISSION OF LTCH.

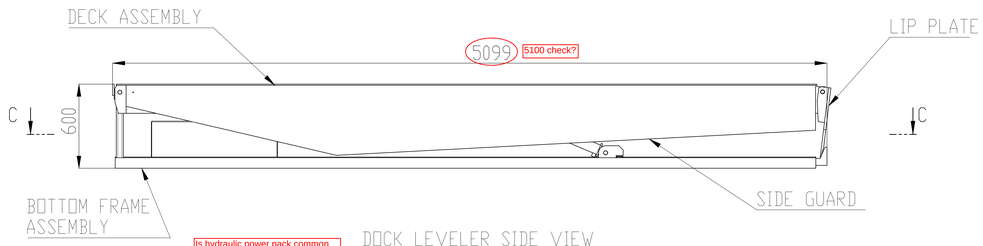
DRG. NO.



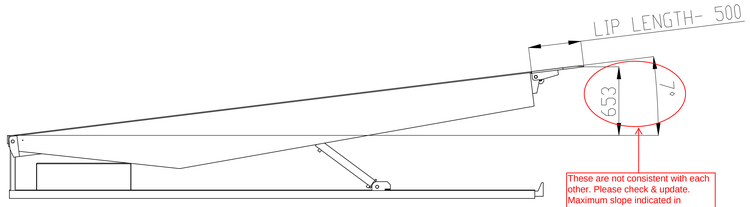
DOCK LEVELER TOP VIEW



ISOMETRIC VIEW OF DOCK LEVELER



DOCK LEVELER SIDE VIEW

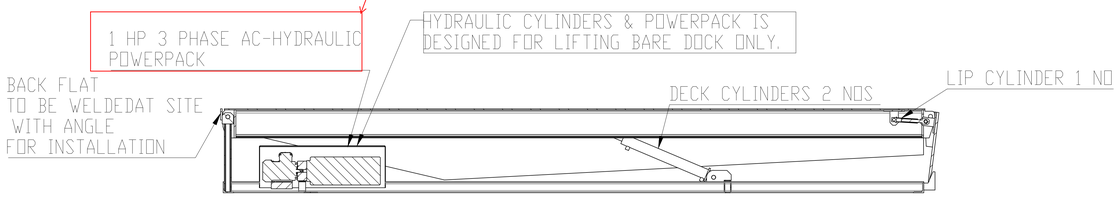


DECK UP HEIGHT - 600 MM



DECK DOWN HEIGHT - 300 MM

Is hydraulic power pack common for both deck cylinders & lip cylinder? If yes, please indicate clearly.

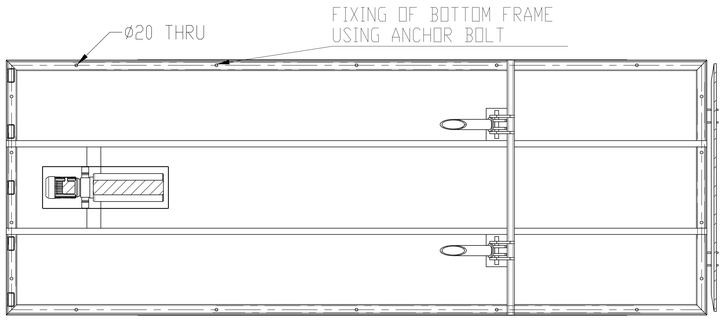


SECTION B-B
SCALE 1 : 15

TECHNICAL SPECIFICATIONS OF PIT MOUNTED ELECRO-HYDRAULIC DOCK LEVELER -

1. SIZE OF DOCK LEVELER = 1820mm(W) X 5000mm ~~(W)~~ (L)
2. OVERALL SIZE OF PRODUCT = 1820mm(W)X5100mm(L)X600mm ~~(H)~~ (D)
3. CAPACITY = 6 TON POINT LOAD
4. LIP PLATE SIZE = 500mm(L)X1820mm(W)
5. DECK PLATE = CHEQUERED SHEET 1820mm(W)x5000mm(L)
6. DECK & LIP UP-DOWN OPERATION BY HYDRAULIC CYLINDER
7. DECK UP HEIGHT=(+) 600 MM
8. DECK DOWN HEIGHT=(-) 300MM
9. HEIGHT OF DOCK LEVELER FROM ROAD LEVEL= 900 mm ,
10. OPERATION CONTROL BY PUSH BUTTONS ON CONTROL PANEL MOUNTED ON STAND NEAR DOCK LEVELER
11. ELECTRICAL SUPPLY OF EQUIPMENT = 415 VOLT 3 PHASE AC , 50HZ
12. PAINT = FIROZI BLUE
13. MATERIAL OF CONSTRUCTION = MILD STEEL AS PER IS STANDARD
14. MANUFACTURING STANDARD OF DOCK LEVELER =EN1398
15. ELECRICAL MOTOR COMPLIANCE ZONE 1-IIC-T3 HAZARDOUS AREA . FOR OTHER EQUIPMENT=PESO CERTIFICATION
16. PIT SIZE = 1870 mm(W)X5100mm(L)x600 mm(D)
17. DOCK LEVELER FIXING IN PIT BY ANCHOR BOLT .
18. REQUIRED PIT & ALL CIVIL WORK FOR MOUNTING OD DOCK LEVELER IS IN ~~CLIENT~~ LTHE SCOPE .
19. REFER PIT DRAWING FOR CONSTRUCTION OF PIT .

Provide reference document number



SECTION C-C
SCALE 1 : 15

EIL Comments:
1. Include a sheet showing both dock levellers together for clarity w.r.t. relative location.
2. Tabulate BOM for one no. dock leveller.
3. "Client" wherever mentioned in the drawing shall be replaced by "LTHE".
4. Other comments as marked.

List here "Civil Drawing for Dock Leveller (504-LZ-930A/B)"

DRAWING NO. | DRAWING TITLE | REV

REFERENCE DOCUMENTS / DRAWINGS

DRG. CATEGORY: (USE TICK MARK) | R | I | EIL REVIEW/APPROVAL CODE: | VPTL NO.:

SCH. DATE OF SUBMISSION:

DOCUMENT STATUS
STATEMENT OF SUB: 1st | 2nd | 3rd
☐ FINAL FOR CODE-1 ☑ AS BUILT

ISSUED FOR
☐ AS BUILT ☐ INFORMATION
☐ CONSTRUCTION ☐ REVIEW

SIGN: DATE:

☐ REVIEW CODE 1 – NO COMMENTS. PROCEED WITH MANUFACTURE / FABRICATION/ CONSTRUCTION AS PER THE DOCUMENT.
☐ REVIEW CODE 2 – PROCEED WITH MANUFACTURE / FABRICATION / CONSTRUCTION AS PER THE COMMENTED DOCUMENT. REVISED DOCUMENT REQUIRED.
☐ REVIEW CODE 3 – DOCUMENT DOES NOT CONFORM TO BASIC REQUIREMENTS AS MARKED. RESUBMIT FOR REVIEW.
☐ R – DOCUMENT IS RETAINED FOR RECORDS. PROCEED WITH MANUFACTURE / FABRICATION
☐ V–VOID

NAME | DISCIPLINE | SIGN. | DATE

REV.	DESCRIPTION	DRN	CHECKED	APPROVED	REVIEWED	APPROVED	DATE
00	ISSUED FOR APPROVAL	SMS	VJ	AK			

OWNER: HINDUSTAN PETROLEUM CORPORATION LIMITED
VISAKH REFINERY MODERNIZATION PROJECT
EPCC-03 PACKAGE FOR RESIDUE UP-GRADATION FACILITY (RUF)

PMC : ENGINEERS INDIA LIMITED (A Govt. of India Undertaking)

LSTK CONTRACTOR: L&T Hydrocarbon Engineering

DETAILED ENGINEERING CENTER: L&T - CHIYODA LIMITED

VENDOR : NIHVA TECHNOLOGIES PVT LTD
VENDOR PO NO: REFINARY/75000-82256/VMM
VENDOR DWG. NO: NIHVA_20-21_0157

DATE: 31.05.21 | SCALE: 1:1 | TITLE: ~~DATASHEET WITH~~ GAD FOR DOCK LEVELER (504-LZ-930A/B)

DEPT: DESIGN

DRN BY: SMS | CHK BY: VJ | APP BY: AK

PROJECT: L&T-RUF | DRG. NO.: B016-RUF-L-T-504-MP-GN-NTPL-(1)-13002 | PWO NO.: | SH.1 OF 1 | REV. 00

CAD REF: DO NOT SCALE DRAWING | FILE:

L&T - CHIYODA LIMITED

☐ REVIEW CODE 1 – NO COMMENTS. PROCEED WITH MANUFACTURE / FABRICATION/ CONSTRUCTION AS PER THE DOCUMENT.
☐ REVIEW CODE 2 – PROCEED WITH MANUFACTURE / FABRICATION /CONSTRUCTION AS PER THE COMMENTED DOCUMENT. REVISED DOCUMENT REQUIRED.
☐ REVIEW CODE 3 – DOCUMENT DOES NOT CONFORM TO BASIC REQUIREMENTS AS MARKED. RESUBMIT FOR REVIEW.
☐ R – DOCUMENT IS RETAINED FOR RECORDS. PROCEED WITH MANUFACTURE / FABRICATION
☐ V–VOID

COMMENTED BY	REVIEWED BY	APPROVED BY	DATE

THIS REVIEW / APPROVAL DOES NOT RELIEVE THE VENDOR / MANUFACTURER OF THE FULL RESPONSIBILITY AS TO DESIGN CORRECTNESS OF DIMENSIONS & COMPLIANCE WITHALL APPLICABLE REGULATIONS AND WITH THE REQUIREMENTS MADE IN PURCHASER SPECIFICATIONS.

DOCUMENT REVIEW DETAILS

1	No Comments.	
2	Proceed with manufacture / fabrication as per the commented document. Revised document required.	
3	Document does not conform to basic requirements as marked. Resubmit for review.	
R	Retained for Records.	V Void.

Signature with Date: 03-SEP-2021
Name of Reviewer: AJEET KUMAR (8580)
Page No.: 13 of 19

DATA SHEET FOR DOCK LEVELLER - (504-LZ-930A/B)

Sr No	Parameter	Remark
1	Required Capacity	06 Ton
2	Leveller operation mode	AC power pack hydraulically operated
3	Lip operation mode	hydraulic
4	Operation control	by push buttons and control panel
5	~~Loading platform~~ height from road Level (of dock leveller)	900 mm
6	Travel above & below dock level	600 mm & 300 mm
7	Platform Size (W x L)	1820 mm x 5000 mm
8	Lip size (W x L)	1820 mm x 500 mm
9	Maximum slope can be achieved	7 Degree

* Please refer Attachment-1 for standard catalog of Dock Leveller.

Both deck up & deck down slopes to be indicated. Also refer comments on GAD in this connection.

DOCUMENT REVIEW DETAILS

1 No Comments. Doc. B016-RUF-LT-504-MP-GN-NTPL(1)-13002 Rev. 00

2 ✓ Proceed with manufacture / fabrication as per the commented document. Revised document required.

3 Document does not conform to basic requirements as marked. Resubmit for review.

R Retained for Records. V Void.

Signature with Date 03-SEP-2021

Name of Reviewer AJEET KUMAR (B565)

Page No. 14 of 19

ALL RIGHTS RESERVED
THIS DOCUMENT AND ANY DATA AND INFORMATION CONTAINED THEREIN ARE CONFIDENTIAL AND THE PROPERTY OF L&T HYDROCARBON ENGINEERING (LTHE) AND THE COPYRIGHT THEREIN IS VESTED IN LTHE NO PART OF THIS DOCUMENT, DATA, OR INFORMATION SHALL BE DISCLOSED TO OTHERS OR REPRODUCED, STORED OR USED FOR ANY PURPOSE WHATSOEVER, EXCEPT WITH THE PRIOR WRITTEN PERMISSION OF LTHE.

ATTACHMENT-1

NIHVA

NIHVA TECHNOLOGIES PVT. LTD.

Dock Leveller



- SOLUTIONS
- SERVICES
- SATISFACTION

DOCUMENT REVIEW DETAILS

1 No Comments. **Doc. B016-RUF-LT-504-MP-GN-NTPL(1)-13002 Rev. 00**

2 ✓ Proceed with manufacture / fabrication as per the commented document. Revised document required.

3 Document does not conform to basic requirements as marked. Resubmit for review.

R Retained for Records. V Void.

Signature with Date **03-SEP-2021**

Name of Reviewer **AJEET KUMAR (B565)**

Page No. **15 of 19**

www.nihva.com

NIHVA

NIHVA TECHNOLOGIES PVT. LTD.

DOCK LEVELLER

LOADING EQUIPMENTS

NIHVA Technologies Pvt.Ltd is the premier solutions provider of Safe, Effective, and technologically innovative material handling equipment's.

We offer verity of Loading Equipments including Dock levellers, Dock Shelters, Dock Seal, Scissor Lifts, Good Lifts along with other material handling products. We offer turnkey solutions for the most demanding freight-handling needs. Dock Levellers are one of the simplest equipment, designed according to the Indian conditions. Continues operations & ruggedness for not only helping the loading & unloading the vehicles easily but also reducing the dispatch time of the vehicles significantly.

Our gamut of hydraulic dock levellers design & fabricated extensive used for heavy loading and unloading.







FEATURES

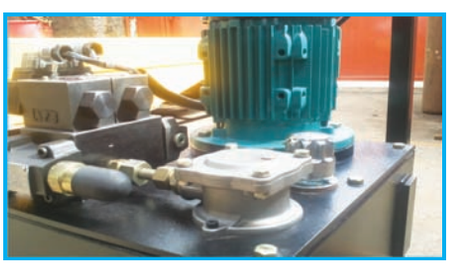
- Robust design, safe operations and long term reliability.
- Total control on basic hydraulic like pressure and flow.
- Velocity safety stops.
- Simple electric panel with push button.
- Priority to safe working operations.
- Long excellent stability.
- Consistency in working performance.

DOCUMENT REVIEW DETAILS
1 No Comments.
2 Proceed with manufacture / fabrication as per the commented document. Revised document required.
3 Document does not conform to design requirements as marked. Resubmit for review.
R Retained for Records.
Signature with Date 03-SEP-2021
Name of Reviewer AJEET KUMAR (B565)
Page No. 16 of 19
Doc. B016-RUF-LT-504-MP-GN-NTPL(1)-13002 Rev. 00



Why NIHVA dock leveller?

- Heavy duty structure to take the dynamic load.
- Stronger Base Frame with self-standing capability.
- Stronger Lip & rear Hinges.
- 16 mm thick lip plate with step at tip for adopting smaller vehicles.
- Rear Strong hinge support for self standing capacity.
- Collapsible Telescopic Side Guard for safety protection.
- Single Approachable steady rest for maintenance.
- Stand alone power pack and control panel for ease of operation and maintenance.







DOCUMENT REVIEW DETAILS

1 No Comments. **Doc. B016-RUF-LT-504-MP-GN-NTPL(1)-13002 Rev. 00**

2 Proceed with manufacture / fabrication as per the commented document. Revised document required.

3 Document returned with comments, resubmit as marked. Resubmit for review.

R Retained for records. V Void.

Signature with Date **03-SEP-2021**

Name of Reviewer **AJEET KUMAR (B565)**

Page No. **17 of 19**

SERVICE BACK UP / MAINTENANCE / AMC



NIHVA SERVICE SOLUTIONS

Preventing a problem is always better than fixing one. That simple philosophy is behind all our Service Contracts as they are designed to reduce operational risks and help provide you with a trouble-free materials handling operation. This allows you to concentrate on your core business rather than worrying about unexpected breakdowns, scheduling maintenance, ordering parts or managing service engineers.

MULTIPLE OF SERVICES & AMC SOLUTIONS

Simply choose the Service Contract you require and let us do the rest:

- **SAFETY INSPECTION**

  Designed to keep your Equipment's operating safely and provide protection for your workforce. This contract ensures that your Equipment's always comply with the safety standards set by national laws and regulating bodies.

- **PREVENTIVE MAINTENANCE**

  This contract is an effective way to prevent unplanned downtime and costly repairs. We take responsibility for maintaining your Equipment according to a pre-determined plan as well as ensuring that they meet all safety requirements.

- **FULL SERVICE**

  This contract is designed to take care of all your equipment's maintenance and repair requirements in order to provide you with the highest levels of operational performance and uptime.

  We offer the services in order to customize your contract as per your exact needs: 24/7 service, On-site NIHVA service technician, entire NIHVA maintenance team along with combined support package.

OUR ACTIVITIES

- Door Automation
- Automation Solutions
- Material Handling Equipments

DOCUMENT REVIEW DETAILS
1 No Comments.
2 Proceed with manufacture / fabrication as per the commented document. Revised document required.
3 Document does not conform to the requirements as marked. Resubmit for review.
R Retained for Records. V Void.
Signature with Date 03-SEP-2021
Name of Reviewer AJEET KUMAR (B565)
Page No. 18 of 19
Doc. B016-RUF-LT-504-MP-GN-NTPL(1)-13002 Rev. 00

OPTIONAL ACCESSORIES

Rubber Bumper



Flashing Light



Traffic Light



Dock Lights



Vehicle Wheel Restraint



Vehicle Wheel Alignment guides



NIHVA TECHNOLOGIES PVT. LTD.

Regd. Office:
8/1 Coromant Hsg. Soc. Sec. No. 27,
Nigdi, Pradhikaran, Pune - 411044.

Tel. : + 91 20 6630 4033
Mob. : + 91 855 109 1333
Email: sales@nihva.com
Web. : www.nihva.com

NIHVA

DOCUMENT REVIEW DETAILS

1 No Comments.
2 Proceed with manufacture/fabrication as per the commented document. Revised document required.
3 Document does not conform basic requirements as marked. Resubmit for review.
R Retained for Records. V Void.

Signature with Date 08-SEP-2021
Name of Reviewer AJEET KUMAR (B565)
Page No. 19 of 19

Doc. B016-RUF-LT-504-MP-GN-NTPL(1)-13002 Rev. 00